Name	Subject Class	Class	Candidate Number
	2BI		

ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2017

BIOLOGY **9744/03**
Long Structured and Free-response Questions **21 Aug 2017**
PAPER 3 **2 hours**

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name, subject class, form class and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions in the spaces provided on the Question Paper.

Section B
Answer any **one** question in the spaces provided on the Writing Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show working or if you do not use appropriate units.

At the end of the examination, fasten your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

FOR EXAMINER'S USE	
1	
2	
3	
4 / 5	
TOTAL	**75**

This document consists of **17** printed pages.

For Examiner's Use

Section A

Answer **all** the questions in this section.

1 Dengue viruses (DENV) are responsible for millions of infections each year in tropical and subtropical areas of the world. According to the World Health Organization, dengue incidence has increased significantly over the past 50 years, turning this infection into the most important mosquito-borne disease in the world and a global health challenge. Fig. 1.1 shows the general global distribution of dengue fever in the year 2005.

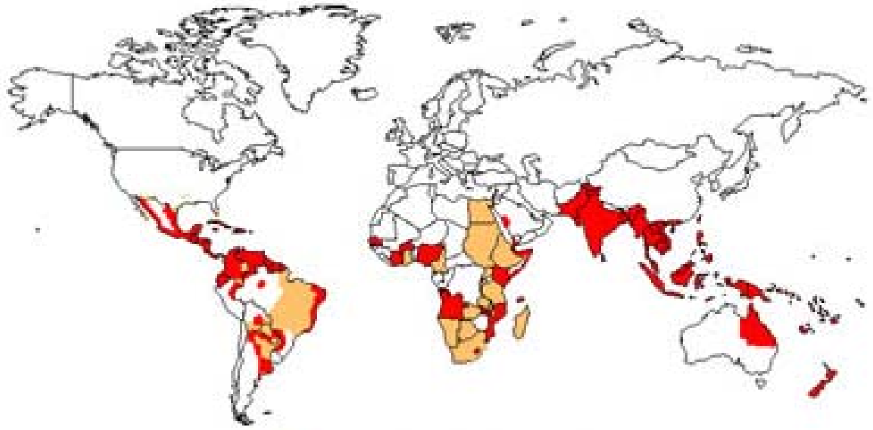

Fig. 1.1

(a) (i) Climate is one of the important factors that affects the distribution of dengue.

Predict and justify the expected distribution of dengue by the end of the 21st century.

1. Geographical range of Aedes mosquito (vector) and dengue will expand towards the two poles/ to areas infested with mosquitoes;
2. As global temperature is predicted to rise (by 4°C)/ global warming/ climate change due to (increase in greenhouse gas emission);
3. resulting in favourable temperatures for mosquito breeding/ suitable breeding places in the temperate regions/ increased viral load/ higher rate of DENV replication;

[3]

For Examiner's Use

(ii) Symptoms of dengue such as fever usually develop within 4 to 7 days after being bitten by an infected mosquito and is often associated with joint pain.

Explain how DENV may cause these symptoms.

1. Virus stimulates/ activates innate/ non-specific immune response/ immune cells (e.g. dendritic cells) in the first four days;
2. Interferons/ cytokines/ histamine released result in inflammation, causing pain;
3. Pyrogen released by activated macrophages leads to rise in systemic body temperature resulting in dengue fever;

***Award mark only when reference is made to respective symptoms**

[3]

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During an infection, DENV will elicit a primary humoral immune response which involves B cell activation as shown in Fig. 1.2.

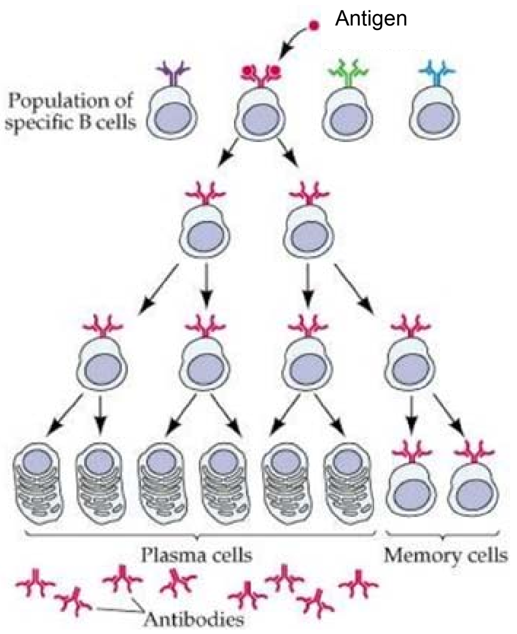

Fig. 1.2

(iii) With reference to Fig. 1.2 and your own knowledge, explain the significance of mitosis in B cell activation.

1.* Clonal expansion occurs via mitosis to produce <u>genetically identical</u> daughter cells;
2. This results in increased/ faster production of antibodies/ memory cells/ plasma cells
3. which can bind to <u>complementary/ specific</u> antigen present;
4. to mediate fast clearance/ destruction of pathogen/ increased rate of phagocytosis/ faster response during secondary infection;
5. Somatic hypermutation during clonal expansion results in antibodies with increased affinity to antigen;

***Compulsory point**

[3]

For Examiner's Use

(iv) Memory B cells have long life spans because they do not actively undergo cell division. However, upon activation, a memory B cell can undergo many rounds of proliferation.

Suggest why activated memory B cells can undergo multiple rounds of cell division without dying.

1. Length of telomeres maintained due to active telomerase/ expression of telomerase upon activation;

R! Length of telomere very long

[1]

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(b) Bacteria has a natural defence system against viral infections involving RNA sequences known as clustered regularly interspaced short palindromic repeats (CRISPR).

When the bacteriophage infects a bacterial cell, the viral genome released into the bacterial cell is cleaved. Subsequently, a cleaved portion of the viral genome is integrated into the bacterial genome. The bacterial cell detects the phage DNA integrated within its genome and produces a type of RNA known as the CRISPR RNA. The CRISPR RNA contains a sequence that is complementary to that of the integrated phage DNA. When the next phage infects the same bacterial cell, the CRISPR RNA will bind to its target sequence in the viral genome and a nuclease is recruited to cut the phage DNA, disabling the invading phage.

The sequence of the CRISPR RNA can be edited and subsequently used to cut any DNA sequence at a precisely chosen site in eukaryotic cells. This gives rise to the possibility of correcting mutations associated with genetic disorders. Fig. 1.3 shows how the sequence of a defective allele can be replaced with the sequence of a normal allele from a donor DNA via homologous recombination using the CRISPR technology.

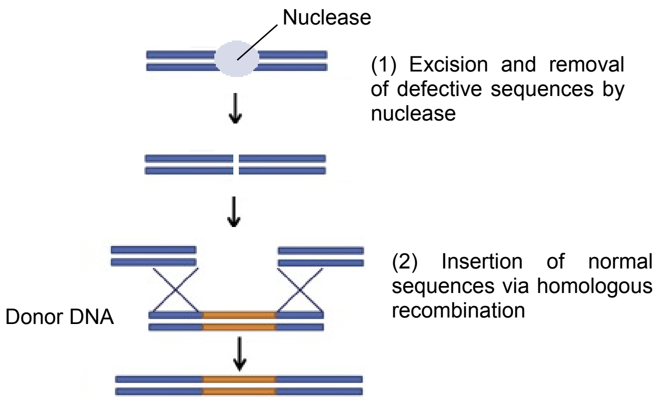

Fig. 1.3

(i) The specificity of CRISPR-mediated immunity in prokaryotes could be applied to eukaryotic cells to make precise changes in the genes of organisms.

Explain why the CRISPR RNA in prokaryotes can also be used in eukaryotic cells.

1. Universal/ similar types of nucleotides (A, U, C, G)/ nucleic acid/ Eukaryotes also contain DNA as its genome; (mark for identify nature of genome)

2. where A is paired with U and C is paired with G via complementary base pairing;

[2]

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(ii) Another technology that can be used to treat genetic disorders is gene therapy. Similar to the CRISPR technology, gene therapy involves the introduction of normal alleles into patients. However, it does not involve the excision and removal of defective alleles. Hence, it can only be used to treat recessive genetic disorder.

With reference to Fig. 1.3, explain the advantages of using CRISPR technology to treat genetic disorders over gene therapy.

1. CRISPR technology can be used to treat both <u>dominant</u> and recessive genetic disorders;

2. as nuclease creates a <u>double stranded break</u> in the DNA to remove the dominant <u>alleles</u> (and gene function is restored when normal alleles are inserted via homologous recombination);
R! dominant genes

3. However, insertion of normal alleles in gene therapy is unable to treat dominant genetic disorders as only a copy of dominant allele is required to show its effect in patients;

4. Normal alleles introduced during gene therapy can be degraded overtime/ temporary/ may not be integrated in the genome vs. long lasting treatment/ one treatment required for CRISPR;

5. CRISPR successfully correct the genetic disorder such that subsequent generations of cells are normal while gene therapy can only affect one generation of cells;

6. Gene therapy has a risk of insertional mutagenesis (if retrovirues are used as a vector) while CRISPR does not have this risk;
7. as in gene therapy, insertion is not specific while in CRISPR, insertion is specific/ precise;

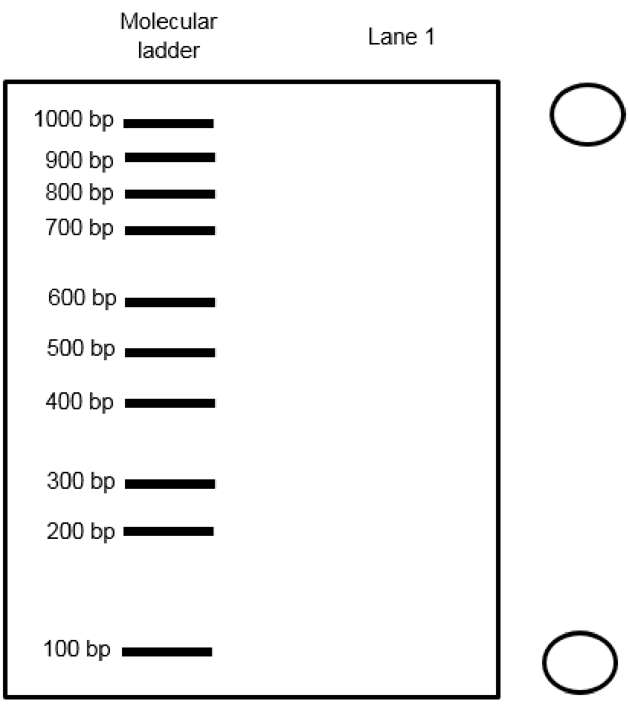
[3]

For Examiner's Use

Polymerase Chain Reaction was used to amplify a gene of 1000 bp. Fig. 1.4 shows the resultant DNA fragment produced. A CRISPR RNA is designed to target a specific sequence found within the gene. The sites of excision by the nuclease are indicated by the arrows. Gel electrophoresis can be used to verify if the target sequence is cut at the precise sites by the nuclease.

Fig. 1.4

(iii) On the electrophoregram below, draw the expected results after the cut fragments are separated if there is specific binding of CRISPR RNA to target sequence in Lane 1. Indicate the charge of the electrodes clearly in the circles on the side of the electrophoregram.



1. A single band at 600, 300, 100 bp position of similar thickness; R! width of band is double
2. Correct labelling of the terminals; [2]

ori

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Fig. 1.5 shows an electrophoregram obtained by a student following excision of the same gene by a nuclease.

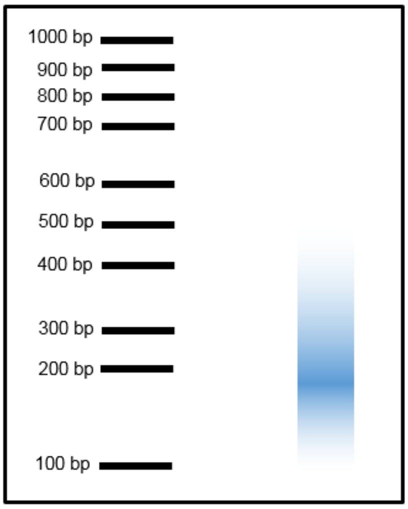

Fig. 1.5

(iv) Suggest how the smear is obtained in Fig. 1.5.

1. Fragments of varied sizes are produced;
2. due to non-specific excision of target sequence;
3. as CRISPR RNA designed is not specific;

[1]

(v) Geneticists attempted to edit the gene responsible for β-thalassaemia, a potentially fatal blood disorder, in human embryos using the CRISPR technology. However, the experiment was terminated prematurely due to a number of unintended mutations and numerous ethical controversies.

Discuss the ethical implications of the application of CRISPR technology on genes in embryos.

1. Permanently changes the genetic makeup of an individual / and future generations;
2. may result in human trait selection/ genetic enhancement/ ref. eugenics by selecting characteristics of offspring;
3. Unintended killing of embryos which is considered a life;
4. Source of embryos may be obtained from donors which did not give their consent;
5. Failure to make informed decisions due to inadequate understanding of the risks associated with technology;
6. Exploitation of women for their oocytes (R! embryos);
7. Experiments using embryos may lead to the killing of embryos and this can desensitize community;

[2]

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(c) One of the three tenets of the cell theory states that all living organisms are composed of one or more cells. Non-cellular life forms such as viruses challenge this tenet as they possess both living and non-living characteristics. The recent discovery of giant viruses, known as Mimiviruses, led scientists to rethink the origin of life and viral evolution. The features of the Mimivirus are as described.

1. It contains a double-stranded DNA genome of approximately 1.2 million base pairs which is significantly larger than the genomes of any other known virus and comparable to a cell.
2. It has some genes which show a high degree of homology to those in bacteria, while some show a high degree of homology to those in eukaryotes.
3. It contains a number of protein-coding genes showing high degree of homology to genes coding for products involved in translation, such as aminoacyl-tRNA synthetases and translation initiation factors. It also contains genes associated with metabolic pathways, DNA repair, and protein folding. However, it is still dependent on its host for translation.

Suggest how Mimiviruses evolved and use the above statements to support your hypotheses.

1. From statement 1: The size and nature of mimivirus genome is similar to a cell;
2. suggesting that it may have evolved from cells;
3. From statement 2: Mimivirus have similar genes/ high degree of homology as genes from as bacteria and eukaryotes;
4. suggesting that mimivirus may acquire genes from cells via horizontal gene transfer;
5. that may have preceded eukaryotes and prokaryotes/ share a common ancestor as eukaryotes and prokaryotes;
5. From statement 3: Mimivirus still depends on the host for translation despite having genes that encodes certain components of translation;
6. suggesting that some of the genes coding for previously complete processes may be lost as it becomes more dependent on its host;

[4]

[Total: 24]

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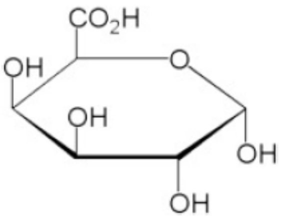
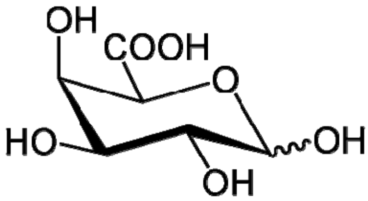
2 Prebiotics is defined as 'an ingredient that results in specific changes in the composition and activity of microbials in the gut thus improving host health'. Prebiotics are non-digestible oligosaccharides that can be selectively fermented by gut bacteria. Currently, most prebiotics are derived from the hydrolysis of simple polysaccharides. Therefore, to develop novel prebiotics, an alternative resource of polysaccharides that supplies oligosaccharides with more diverse and complex structures is required. One of these polysaccharides is pectin.

Fig. 2.1 shows the schematic diagram and natural occurring configuration of D-galacturonic acid, the monomer of pectin. The structure of pectin, a polysaccharide is shown in Fig. 2.2.

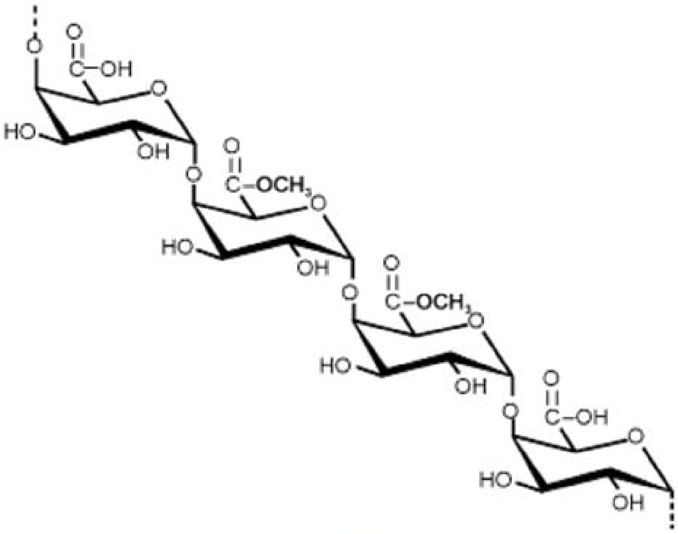


Schematic diagram



Natural occuring configuration

Fig. 2.1



Pectin

Fig. 2.2

For Examiner's Use

(a) With reference to Fig. 2.1 and 2.2, state two structural differences between cellulose and pectin.

1. **Alternating cellulose monomers inverted but pectin monomers all in same orientation;**
2. **D-galacturonic acid monomer in pectin vs beta glucose in cellulose;**
3. **Beta configuration of glucose monomers in cellulose vs alpha configuration of D-galacturonic acid monomer in pectin/ OH group on C1/anomeric carbon below the plane in pectin but in cellulose it is above the plane;**
4. **β (1→4) glycosidic bonds in cellulose while α (1→4) glycosidic bonds in pectin;**
5. **Monomers in pectin are esterified/modified to carry methyl groups but not in cellulose;**

[2]

Rhamnogalacturonan I (RG I) is a type of pectin present in the plant cell wall. The schematic diagram of RG I found in plant cell walls is shown in Fig. 2.3. 'Ac' represents acetyl groups added to RG I.

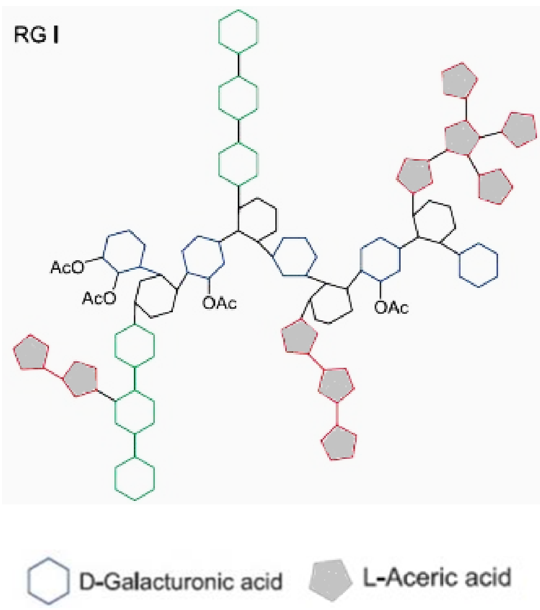


Fig. 2.3

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(b) With reference to Fig. 2.3 and your knowledge of carbohydrates, suggest how the structure of monosaccharides allow for complexity in the pectin structure.

1. **There are different number of C in the ring which gives rise to diversity in monomers used in pectin;**
2. **e.g. L- Aceric acid, D-Galacturonic acid;**
3. **Each ring has multiple OH groups at different/ multiple C position enable the formation of multiple bonds;**
4. **for branching in pectin;**
5. **A variety of monomers can be attached to a single monomer, allowing for diversity and complexity;**
6. **The monomers structure also allow for acetylation, allowing for diversity and complexity;**

[2]

When pectin is consumed by humans, it is digested into oligosaccharides. These oligosaccharides are used by gut bacteria in the process of anaerobic respiration.

(c) Explain why a small yield of ATP can still be achieved with anaerobic respiration.

1. **Glycolysis continues for the synthesis of some ATP via substrate-level phosphorylation;**
2. **this is achieved by regenerating NAD^+ from the reduced NAD produced;**
3. **Pyruvate accepts the hydrogen atoms from reduced NAD and is reduced to lactate (which contains a lot of trapped energy);**
4. **1 molecule of glucose is oxidised/ converted to (2 molecules of) pyruvate with the net yield of 2 ATP (and 2 reduced NAD);**

[3]

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In the presence of oxygen, both glucose and triglycerides can be oxidised to yield large amounts of ATP. Fig. 2.4 shows some steps involved in respiration of triglycerides. Respiration of triglycerides involves the hydrolysis of triglycerides into fatty acid chains which are then broken down into acetyl CoA molecules. These acetyl CoA molecules will then enter the Krebs cycle.

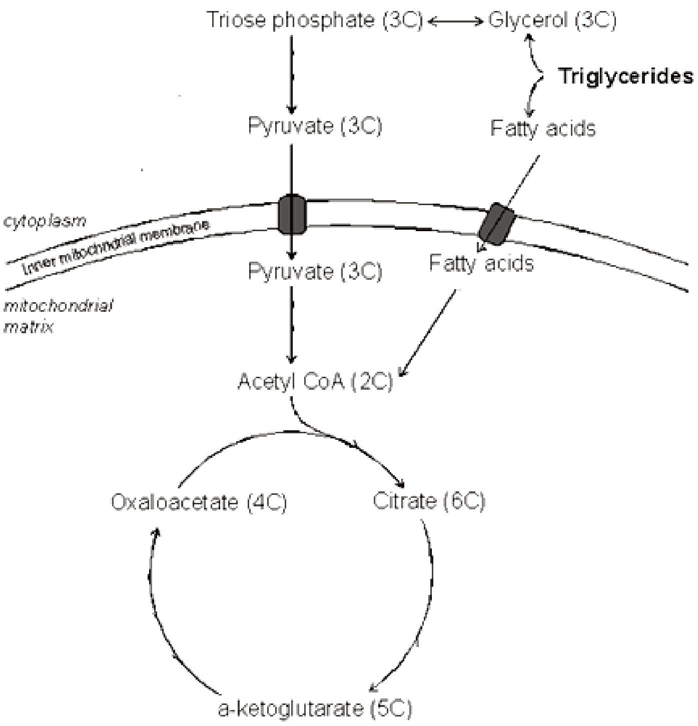

Fig. 2.4

For Examiner's Use

(d) (i) Using the information from Fig. 2.4, calculate the total ATP yield from the complete oxidation of a 14-carbon fatty acid chain. Each NAD yields 2.5 molecules of ATP and each FAD yields 1.5 molecules of ATP. Show your calculation and answer in the space provided.

14-carbon fatty acid chain will yield 7 x 2 carbon chains (7 cycles);
No. of ATP from a 2 carbon-chain = 3 x 2.5 + 1.5 + 1 = 10;
Hence, total ATP yield will produce 7 x 10 ATP molecules = 70 ATP;

[3]

(ii) Explain why fatty acid chains can be completely oxidised only under aerobic conditions.

1. **Fatty acids need to be converted first to acetyl CoA;**
2. **that is only oxidised in the Krebs cycle/ bypasses glycolysis and the link reaction;**
3. **In order to be completely oxidised, <u>oxygen</u> is required as the <u>final electron and proton acceptor</u> in the process of oxidative phosphorylation;**

[3]

[Total: 13]

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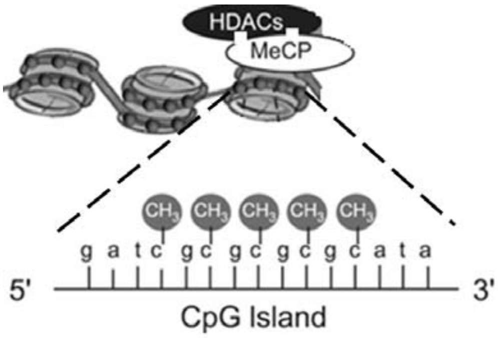
3 CpG islands are regions of DNA where a cytosine nucleotide is followed by a guanine nucleotide in the linear 5' to 3' sequence. CpG islands are typically 300 to 3000 base pairs in length. These CpG islands have been found to be in or near approximately 40% of promoters of mammalian genes. Humans have a higher percentage of promoters with high CpG content.



HDAC: Histone Deacetylase
MeCP: Methyl-CpG-binding protein

Fig. 3.1

Commented [WU1]: Make the protein binding more complementary

(a) **(i)** With reference to the information provided, explain the significance of the presence of CpG islands in many of the genes in the mammalian genome.

1. **CpG islands are sites of DNA methylation;**
2. **Addition of methyl group changes the 3D configuration of the promoter;**
3. **DNA methylation at/near promoter recruits Methyl-CpG-binding protein which in turn recruits and binds to histone deacetylase;**
4. **Complementary binding of Methyl-CpG-binding protein to Histone Deacetylase;**
5. **Histone Deacetylase removes acetyl group from histone restoring positive charge (on lysine tail);**
6. **causes condensation of chromatin;**
7. **preventing access of RNA pol and <u>general</u> transcription factor to/ formation of TIC at promoters;**
8. ***Allows specific genes to be silenced during regulation/ switched on when needed;**

***Compulsory**

[4]

McGhee and Ginder conducted an experiment that examined the effects of inhibiting methylation on gene expression. The experiment was performed using 5-azacytidine in mouse cells. 5-azacytidine is one of many chemical analogs that are structurally similar to the nucleoside cytidine from which cytosine is formed. When these analogs are integrated into growing DNA strands, some, including 5-azacytidine, severely inhibit the action of the DNA methyltransferase enzymes that normally methylate DNA. Interestingly, other analogs, like Ara-C, do not negatively impact methylation.

Scientists hypothesized that if they inhibited methylation by flooding cellular DNA with 5-azacytidine, then they could compare cells before and after treatment to see what impact the loss of methylation had on gene expression. The amount of methylation measured in each treatment is relative to the control. The results are shown in Table 3.1.

Table 3.1

Chemical Added	Number of Differentiated Cells	Amount of Methylation Measured
cytidine (control)	0	100%
Ara-C	0	127%
5-azacytidine	22141	33%

(ii) Explain what the experimental results show about the effects of methylation on gene expression.

1. **The decrease in methylation resulted in an increase in the number of differentiated cells;**
2. **Inhibition by 5-azacytidine decrease amount of methylation from 100 to 33% leads to increase the number of differentiated cell from 0 to 22141 cells;**
3. **While the presence/ increase in methylation of 127% due to Ara-C resulted in 0 differentiated cell OR While the presence of 100% methylation in control resulted in 0 differentiated cell;**

Max 2

4. **Methylation (of CpG region at promoter) causes condensation of chromatin, preventing access of RNA pol (and general TF) to promoter/ Demethylation of genes causes unpacking of chromatin, allowing access of RNA pol to promoter;**
5. **This allows for gene expression leading to expression of specific proteins in the process of differentiation / shows that specific gene involved in differentiation are methylated;**

[4]

(iii) Explain how confidence in the experimental results could be increased.

1. **Repeat experiment to increase reliability of results;**

2. (Add another chemical which will) induce 0% methylation as a positive control/ to ensure that methylation arises are due to the effect of Ara-C or 5-azacytidine; [1]

3. AVP

For Examiner's Use

For Examiner's Use

(b) In 2010, a 10 year old boy with a damaged trachea (windpipe) was given a trachea transplant. A donor trachea was obtained and enzymes were used to remove all the cells, leaving only the collagen as shown in Fig. 3.2.

Fig. 3.2

Some bone marrow was removed from the boys' pelvis and about 2.5 million stem cells were isolated from this bone marrow. These stem cells were then treated with chemicals to stimulate proliferation and injected into the collagen.

The collagen, with the injected stem cells, was then immediately used to replace damaged trachea in the boy. Over a period of time, a fully functioning trachea was formed.

Suggest how the properties of the bone marrow stem cells allow for the formation of a fully functioning trachea.

1. **Bone marrow stem cells are multipotent stem cells;**
2. **They are able to undergo indefinite mitosis due to active telomerase;**
3. **To form/ maintain a constant pool of stem cells for differentiation;**
4. **Bone marrow stem cell are undifferentiated cells;**
5. **Which can respond to different environmental factors to change the expression of the bone marrow stem cell into cells in the trachea;**

[4]

[Total: 13]

For Examiner's Use

Section B

Answer **one** question in this section.

Write your answers to this question on the separate writing paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

4 **(a)** Using two named examples, explain the importance of bonds to the function of two different classes of biomolecules. [13]

(b) With reference to named examples, describe the range of roles performed by lipids in living organisms. [12]

[Total: 25]

5 **(a)** Outline the processes that result in genetic variation in nature and explain the significance of such processes. [13]

(b) "The endomembrane system is critical in the synthesis of proteins". Discuss. [12]

[Total: 25]

For Examiner's Use

Essay Question 4a

Using two named examples, explain the importance of bonds to the function of two different classes of biomolecules. [13]

<u>Nucleic Acid (e.g. DNA/ RNA)</u>

1. **The role of DNA is to store information;**
2. **and pass it on / transmit information from one generation to the next;**
3. **To ensure that DNA is stable;**
4. ***numerous <u>hydrogen bonds</u> can found between the nitrogenous bases;**
5. ***Strong covalent bonds e.g. <u>phosphodiester bonds</u> between adjacent nucleotides in sugar-phosphate backbone;**
6. **<u>*Hydrophobic interactions</u> between <u>stacked</u> nitrogenous bases;**
7. **Weak hydrogen bonds between nitrogenous bases can be easily broken to allow separation of DNA;**
8. **for DNA replication, transcription and DNA repair (at least 1);**
9. **Complementary base pairing by hydrogen bonds;**
10. **allow for accurate transmission of information;**
11. **The sugar phosphate backbone that is negatively charged allows association with positively charged histone via <u>ionic interactions;</u>**
12. **to allow for condensation;**
13. **Hence, preventing the breakage of DNA/ lost of genetic information during nuclear division;**
14. **Condensation also allows for the DNA to be packaged in the nucleus of the cell;**

<u>Protein – Enzyme/ antibody</u>

15. ***Amino acids join together by (strong covalent bonds known as) <u>peptide bonds</u>;**
16. **The presence of hydrogen bonds formed between the –CO and –NH groups of the polypeptide backbone to form α-helices and B pleated sheets;**
17. **The 3D conformation of the protein is maintained by ionic, hydrogen bonds and hydrophobic interactions and disulfide bonds between R-groups (at least 2 bonds);**
18. **For proteins with quaternary structure, the 3D conformation of the protein is maintained by ionic, hydrogen bonds and hydrophobic interactions and disulfide bonds between R groups of different polypeptide chain (at least 2 bonds);**
19. **This allow formation of active site complementary to substrate;**
20. **for specificity in catalysis;**
21. **This allow formation of binding sites/ sites other than the active sites complementary to activator/ inhibitor/ cofactors;**
22. **For regulation of reactions;**
23. **R groups of binding amino acid residue form transient bonds with substrate;**
24. **Allow formation of enzyme-substrate complex;**

For Examiner's Use

Protein - Collagen

25. *Amino acids join together by (strong covalent bonds known as) peptide bonds (to form the alpha chain);
26. which has a repeating tri-peptide sequence of glycine-X-Y;
27. where X is often proline and Y is often hydroxyproline or hydroxylysine;
28. Polymerisation allows for the formation of a long molecule that allows collagen to function as a structural protein;
29. Collagen is insoluble in water due to absence of hydrogen bonds with water;
30. As glycine and proline are hydrophobic;
31. Collagen has high tensile strength;
32. *Due to covalent cross-links between different tropocollagen molecules;
33. *Cross linking by hydrogen bonds between alpha chains/ within triple helix/ tropocollagen (R! within collagen); (award mark if linked to insolubility and tensile strength);
34. Bundling for the formation of a fibril and subsequently fibres;

Protein - Haemoglobin

35. Amino acids join together by (strong covalent bonds known as) peptide bonds (to form the alpha and beta chains which are 141 amino acid and 146 amino acid long);
36. The presence of hydrogen bonds formed between the –CO and –NH groups of the polypeptide backbone to form α-helices;
37. The 3D conformation of the protein is maintained by ionic, hydrogen bonds and hydrophobic interactions between R-groups (at least 2 bonds);
38. (Relatively weaker) ionic bonds/ and hydrogen bonds occur between dimer pairs (in the deoxygenated state);
39. (Relatively stronger) hydrophobic interactions/ and hydrogen bonds between α chain and β chain form (stable) αβ dimers;
40. This allows for the arrangement of hydrophobic amino acid residues within the interior of the globular structure;
41. allowing for the formation of a haem binding pocket/ hydrophobic environment / deep hydrophobic cleft for the haem group to bind to oxygen;
42. Hydrophilic amino acid residues are found at the surface of the globin;
43. Allowing hydrogen bonds to be formed between the water and hydrophilic amino acid residues;
44. Allows for solubility in a aqueous medium/ allows it to be a good transporter of oxygen in blood;
45. The hydrogen bonds between the two dimers allow for cooperativity to occur;
46. Increase rate of loading or unloading of oxygen on/ off Hb;

For Examiner's Use

Protein - G-Protein Linked Couple Receptor

47. *Amino acids join together by (strong covalent bonds known as) peptide bonds (to form the alpha chain);
48. *The presence of hydrogen bonds formed between the –CO and –NH groups of the polypeptide backbone to form 7 α-helices;
49. *The 3D conformation of the protein is maintained by ionic, disulfide, hydrogen bonds and hydrophobic interactions between R groups (at least 2 bonds);
50. *This allows for the arrangement of hydrophobic/ non-polar R groups of amino acids facing exterior of α helices to interact with hydrophobic/ hydrocarbon tails of the phospholipid bilayer;
51. hence allowing for embedment of the protein in the membrane;
52. This also allows for the formation of a binding site to allow binding of a signal molecule/ ligand;
53. And the formation of a binding site to allow binding of the G protein;
54. Hence allowing for the signal molecule to bind and trigger conformation change in GPLR;
55. And activated GPLR can bind to the G protein and activate G protein when GTP displaces GDP;

Mention 3 out of 4 compulsory points

Carbohydrates: Starch and/ or glycogen

56. Glycosidic bonds (are strong covalent bonds that) join many α-glucose monomers together to form starch / glycogen;
57. Which can be released upon hydrolysis as respiratory substrates;
58. This also forms a large molecule so that the molecule is insoluble in water;
59. *α-1,4-glycosidic bonds forms a helical structure;
60. which causes the molecule to be compact for storage;
61. Hydroxyl groups of glucose residues that project into the interior of the helices;
62. Hence, absence of hydrogen bonds with water causing starch / glycogen to be insoluble in water;
63. Hence, they can be stored in large quantities without affecting the osmotic potential of cells;
64. *α-1,6-glycosidic bonds allows for amylopectin / glycogen to be highly branched;
65. Hence, a greater amount of carbohydrates can be stored per unit volume;
66. It also allows for many enzymes to act on it at the same time;
67. Allows for quick release of glucose (for an increased rate of respiration);

For Examiner's Use

<u>Carbohydrates - Cellulose</u>

68. <u>Glycosidic bonds</u> (are strong covalent bonds) join many β-glucose monomers together to form cellulose;
69. This also forms a large molecule so that the molecule is insoluble in water;
70. *B 1,4 glycosidic bonds allow the formation of straight chains;
71. <u>*Hydrogen bond cross links</u> between hydroxyl groups of adjacent chains
72. prevent the hydroxyl groups from forming hydrogen bonds with water hence allowing it to be insoluble in water;
73. Also allows for formation of microfibrils and macrofibrils
74. which allows cellulose to have tremendous tensile strength;
75. Allowing it to perform its function as a structural molecule;
76. To help prevent cells from bursting / maintain shape of cell / allows for cell turgidity;

<u>Lipid - Triglycerides</u>

77. Ester bonds allow for the joining of three fatty acids and one glycerol group;
78. Presence of long hydrocarbon chain results in a hydrophobic molecule that is insoluble in water;
79. The presence of multiple energy rich C-H bonds;
80. Great amount of energy to be released during oxidisation during respiration to produce ATP;
81. Triglycerides contain much more energy per gram than either carbohydrates / proteins i.e. 1 gram of fat respired produces twice as much ATP as 1 gram of carbohydrate or protein;
82. Triglycerides can be compacted together for storage;
83. via hydrophobic interactions between the hydrophobic hydrocarbon tails;

<u>Lipid - Phospholipids</u>

84. *Ester bonds join two fatty acids, one phosphate group and one glycerol together to
85. *Phosphate head is negatively charged and hydrocarbon chains are non-polar;
86. Hence giving rise to its <u>amphipathic</u> nature;
87. *Hydrogen bonds formed between the phosphate head and water/ aqueous medium;
88. *and hydrophobic interactions between the hydrophobic hydrocarbon tails;
89. Allow for the formation of cell membrane with phospholipid bilayer;
90. Prevent polar molecules/ ions to pass through the hydrophobic core/ allows for regulation of specific molecules/ions to move into and out of the cell interior;
91. The presence of C=C bonds/ unsaturated hydrocarbon tails causes <u>kink</u>;
92. cannot pack so closely together;
93. Increase the fluidity of the cell membrane at low temperatures;
94. Phospholipids with saturated hydrocarbon tails/tails without C=C bonds;
95. can pack together closely;
96. so that membrane remains stable at high temperatures;

Mention 3 out of 4 compulsory points

For Examiner's Use

<u>Lipid derivative – Cholesterol (to be marked under lipids)</u>

97. Covalent bonds allow the formation of cholesterol which consists of a carbon skeleton with four fused rings and hydroxyl group at one end;
98. *Ionic bonds/ hydrogen bonds between the hydroxyl group on cholesterol interacts with the polar phosphate group of membrane phospholipids;
99. *Hydrophobic interactions between cholesterol and hydrocarbon chain;
100. Allows it to be embedded in the membrane;
101. Hence, at relatively warm temperatures, cholesterol makes the membrane less fluid;
102. by restraining the movement of phospholipids;
103. At low temperatures, cholesterol prevent solidification;
104. by disrupting the regular packing of phospholipids;

QWC: Candidates should discuss points from two different categories of biomolecules with an example each with a coherent and logical flow;

Max 7m per category

Mark first 2 named examples if more than 2 named examples are given

To include antibodies, general proteins

Essay Question 4b

With reference to named examples, describe the range of roles performed by lipids in living organisms. [12]

<u>**Triglycerides**</u>

1. Serve as long term energy store;
2. As there are many energy rich C-H bonds which can be oxidized;
3. during respiration to produce ATP;
4. They contain much more energy per gram than carbohydrates or proteins/ higher calorific value;
5. They are only oxidised after carbohydrates are depleted;
6. Especially important for hibernating animals;
Max 2m for pt 1 - 6
7. Serve as excellent heat insulator underneath the skin;
8. prevent excessive loss of heat;
9. important for aquatic mammals and mammals living in cold climates;
10. Provide buoyancy;
11. because they are less dense compared to water;
12. this is important for aquatic animals.;
13. Provide mechanical protection because they are found around vital organs;
14. and helps to cushion them against physical trauma and impact;
15. Serve as a source of metabolic water;
16. During respiration, the same mass of triglycerides releases twice as much water as carbohydrates;
17. this is important for desert mammals;
18. Solvent for fat-soluble vitamins;
19. For absorption and storage of vitamins A, D, E and K in the body;
Max 4m for pt 7 - 19

<u>**Phospholipids**</u>

20. Amphipathic nature allows them to form a phospholipid bilayer;
21. which is the main component of the cell membranes/ named example of membrane e.g. cell surface membrane, thylakoid, nuclear membrane;
22. This allows for regulation of specific molecules/ions to move into and out of the cell interior;
23. The degree of saturation in fatty acid chains regulates fluidity of cell membrane;
24. Phospholipids with saturated hydrocarbon tails;
25. ensure that membrane does not become too fluid at high temperatures;
26. Phospholipids with unsaturated hydrocarbon tails/ C=C bonds in the tails form <u>kinks;</u>
27. to ensure that membrane remains fluid even at low temperatures;
28. Fluid nature of phospholipid membrane also allows for embedding of membrane proteins/ named example of membrane protein and roles e.g. ATP synthase for ATP synthesis;
29. Sphingomyelin, a type of phospholipid is found abundantly in the myelin sheath of neurons;
30. facilitates rapid conduction of nerve impulses;
31. Oligosaccharides can associate with phospholipids to form glycolipids;
32. Which in turn are involved in cell-cell recognition;

For Examiner's Use

Cholesterol

33. **Cholesterol helps to maintain the fluidity of cell membranes;**
34. **At warm temperatures, it makes the membrane less fluid by restraining the movement of phospholipids**
35. **At low temperatures, it hinders solidification by disrupting the regular packing of phospholipids;**
36. **It is a precursor for the synthesis of other steroids e.g. sex hormones such as testosterone and oestrogen;**
37. **It is a precursor for the synthesis of bile salts which aids in the digestion of fats;**

QWC: Candidates should discuss points from at least 2 examples of lipids to cover a range of functions in a coherent flow;

Max 6m for each categories

For Examiner's Use

Essay Question 5a
Outline the processes that results in genetic variation in nature and explain the significance of such processes. [13]

Mutation

1. **Mutation results in genetic variation;**
2. **Mutation can be brought about by errors in DNA replication due to DNA polymerase;**
3. **Mutation can be brought about by environmental factors / chemical agents;**

Sexual Reproduction

4. **Crossing over between non sister chromatids of homologous chromosomes;**
5. **Where homologous chromosomes pairs up to form a bivalent;**
6. **during prophase I;**
7. **Results in different combination of alleles in gametes;**
8. **Independent assortment of homologous chromosomes during metaphase;**
9. **Results in different combination of paternal and maternal chromosomes;**
10. **Random fusion of haploid gametes during fertilization;**

Significance (for mutation and sexual reproduction)

11. **These increases genetic variation in the gene pool of subsequent generation;**
12. **for natural selection to take place;**
13. **where individuals with advantageous traits to be selected for;**

Immune system

14. **Somatic/ VDJ recombination occurs on the B cell (and T cell) receptors genes/ antibody genes/ in undifferentiated B (and T cells);**
15. **Random selection and joining of 1 V, (1 D,) 1 J segments in the light/ heavy chain;**
16. **result in a variety of different variable regions/ antigen binding site;**

Significance

17. **Brings about a large variation/ vast repertoire of B (and T cell) receptors/ antibodies;**
18. **Allows immune cells to recognise a vast repertoire of antigens found on pathogens;**
19. **for adaptive immune response/ destruction and clearance of pathogen;**

20. **Somatic hypermutation occurs in antibody gene coding for V region of activated B cells;**

Significance

21. **Allows for the production antibodies / antigen receptors with higher affinity to antigen;**

22. **Class switching occurs in antibody gene coding for c region/ heavy chain in activated B cell;**

Significance (for immune system)

23. **Allow it to activate different effector cell for immune response;**

For Examiner's Use

Horizontal gene transfer

24. Genetic variation also arises due to <u>horizontal gene transfer</u> in bacteria;
25. <u>Transformation</u> occurs where competent bacteria takes up foreign naked DNA from the environment;
26. *Homologous regions undergo genetic recombination via crossing over/ homologous recombination;
27. Bacteriophage transfer DNA fragment from host to recipient bacteria via <u>specialised</u> due to error in packing of bacteria DNA;
28. or <u>generalised transduction</u> due to excision of bacteria DNA;
29. *Homologous regions undergo genetic recombination via crossing over/ homologous recombination;
30. In <u>conjugation</u>, plasmid DNA is transferred;
31. from F^+ cell to F^- cell through a <u>mating bridge/ conjugation tube</u> (R! sex pilus);

Significance (for horizontal gene transfer)

32. These allows bacteria to gain new genes for metabolism / synthesis amino acid;
33. Allows for evolution into different strains via natural selection;

Mutation and recombination in viruses

34. In viruses, mutation can result in <u>antigenic drift</u>;
35. In viruses, recombination occurs when two or more strains infected the same host cell result in <u>antigenic shift</u>;

Significance (mutation and recombination in viruses)

36. Allows viruses to evade the host immune response;
37. For antigenic shift: and infect new host cells;

QWC: At least 2 categories, processes linked coherently to the appropriate significance;

***Mark once for pt 26 and 29**
Max 7m for each categories (min 1 significance in each category)

For Examiner's Use

Essay Question 5b
"The endomembrane system is critical in the synthesis of proteins" Discuss. [12]

1. **The endomembrane system is confers several benefits in the synthesis of protein;**
2. **In the eukaryotes, endomembrane systems such as the nucleus, rough endoplasmic reticulum, Golgi apparatus (at least 2) are present;**
3. **The nucleus is an organelle where the genetic material in the form of DNA are surrounded by the nuclear envelope;**
4. **This nuclear envelope protects the genetic material from degradation by nuclease present in the cytoplasm;**
5. **Hence maintaining the integrity of the DNA / prevent mutation caused by oxidative agents;**
6. **Nuclear envelope also allows for regulation of protein synthesis at the post transcriptional level;**
7. **via the export of the mRNA to the cytoplasm;**
8. **Rough endomembrane reticulum provides a large surface area for the attachment of ribosomes for the translation of mRNA to polypeptide;**
9. **RER provides an optimal condition for the folding of the polypeptide/ post translational chemical modification by enzymes;**
10. **GA lumen provides an optimal condition for post translational chemical modification by enzymes;**
11. **It also provides an optimal conditions for the enzymatic reactions in post translational chemical modification;**
12. **Increasing the rate of reaction due to compartmentalization which increases the local concentration of enzyme and substrate:**
13. **Hence the RER and the GA allows for the synthesis of proteins that are complexed with carbohydrates / lipids/ producing glycolipids and glycoproteins;**
14. **It also allow the synthesis of lysosomes, membrane bound organelles that contains hydrolytic enzymes;**
15. **involved in autophagy/ intra-cellular digestion/ apoptosis;**
16. **It also allow synthesis of membrane bound proteins;**

17. **However the endomembrane system is not necessary in the synthesis of intracellular proteins;**
18. **mRNA of intracellular proteins are translated by free ribosomes in the cytoplasm;**
19. **which are in turned folded in the cytoplasm into their native conformation (by with the help of chaperone proteins);**
20. **The endomembrane system is also not necessary in the synthesis of protein in the bacteria;**
21. **Membrane bound proteins/ lactose permease in bacteria can be synthesized despite the absence of endomembrane system;**
22. **Bacteria do not have membrane bound organelles;**
23. **(As the genetic material of the bacteria are not bounded by the nuclear envelope,) transcription of the gene and translation of the mRNA can occur simultaneously;**

24. **However, in the absence of nuclear envelope, genes in the bacteria are also subjected to a higher rate of mutation;**
25. **Regulation of protein synthesis occurs predominantly at the transcriptional level;**

QWC: Good spread of knowledge communicated from both sides;

Pt 1 to 16, 24 and 25: max 7
Pt 17 to 23: max 5